

IDEATION PHASE - Brainstorm & Idea Prioritization Template

Project Name: OptiCrop: Smart Agricultural Production Optimization Engine

Date: _____ Team ID: _____

Step 1: Team Gathering, Collaboration and Select the Problem Statement

The team discussed major agricultural challenges including crop selection, soil fertility, weather uncertainty, irrigation and fertilizer management. After evaluating feasibility, innovation and impact, the team selected OptiCrop to provide intelligent crop recommendations using soil and environmental parameters.

Problem Statement: Farmers often select crops without scientific analysis of soil nutrients and climatic conditions, resulting in lower yield and resource wastage. OptiCrop recommends the most suitable crop using N, P, K, pH, temperature, humidity and rainfall.

Step 2: Brainstorm, Idea Listing and Grouping

Idea	Category
Crop Recommendation	Core
Fertilizer Recommendation	Soil
Weather Analysis	Environment
Crop Yield Prediction	Analytics
Disease Prediction	AI
Smart Irrigation	IoT
Farmer Dashboard	Application

Grouped Ideas

- Crop Intelligence: Crop Recommendation, Yield Prediction
- Soil Management: Soil Analysis, Fertilizer Recommendation
- Environmental Monitoring: Weather Analysis, Smart Irrigation
- Farmer Support: Dashboard & Reports

Step 3: Idea Prioritization

Idea	Innovation	Feasibility	Impact	Priority
Crop Recommendation	High	High	High	★★★★★
Fertilizer Recommendation	High	High	High	★★★★

Yield Prediction	High	Medium	High	★★★★
Disease Prediction	Medium	Medium	Medium	★★★

Selected Idea: OptiCrop was prioritized because it provides high social impact, practical implementation, and supports sustainable agriculture through data-driven crop recommendations.

Expected Outcomes

- Accurate crop recommendations
- Improved yield
- Efficient fertilizer usage
- Resource optimization
- Support sustainable farming